

6.5 How can programmable devices and smart technologies provide functionality in system design?

- a. Demonstrate an understanding of how smart materials change the functionality of engineered products, such as:
- colour changes, shape-shifting, motion control, self-cleaning and self-healing.



- b. Demonstrate an understanding of how programmable devices are used to add functionality to products, relating to coding of and specific applications of programmable components, such as:
- how they incorporate enhanced features that can improve the user experience and solve problems in system design
 - how they use basic techniques for measuring, controlling, storing data and displaying information in practical situations
 - electronic prototyping platforms and integrated development environments (IDE) for simulation in virtual environments
 - the use of programmable components and microcontrollers found in products and systems, such as robotic arms or cars
 - creating flowcharts to describe processes and decisions within a process to control input and output components.



7. Manufacturing processes and techniques

Considerations

*Maths &
Science*